

MATH 199/249: INTRODUCTION TO GENERAL TOPOLOGY
PROBLEM SET #1

FALL 2026

Due Thursday, September 24. Cite all sources, including people, books, papers, websites, and AI tools. Write up your answers in your own words.

Problem 1. Let z_1 and z_2 be complex numbers.

- (1) Prove that $|z_1 + z_2| \leq |z_1| + |z_2|$.
- (2) Prove that $||z_1| - |z_2|| \leq |z_1 - z_2|$.

Hint: Both parts can be solved using the triangle inequality.

Problem 2. Let $d, e: \mathbf{R}^2 \rightarrow \mathbf{R}$ be defined by

$$d(\vec{u}, \vec{v}) = \max\{|v_1 - u_1|, |v_2 - u_2|\},$$
$$e(\vec{u}, \vec{v}) = \sqrt{|v_1 - u_1|^2 + 4|v_2 - u_2|^2}.$$

Decide which of these functions is a metric. Justify your answer. For each function that is a metric, sketch its unit circle at the origin.

Problem 3. For all positive integers n , let

$$a_n = \sqrt{n+1} - \sqrt{n} \quad \text{and} \quad b_n = \frac{1}{n}(\sqrt{n+1} - \sqrt{n}).$$

Decide which of these sequences converges, and if so, to which limit(s). Justify your answer.

Problem 4. Let $p_n = \frac{1}{n}$ and $q_n = \sum_{k=1}^n p_k$.

- (1) Show that the sequence $\{p_n\}_{n=1}^\infty$ converges. What is its limit?
- (2) Using induction on positive integers m , show that

$$\text{if } N = \sum_{j=0}^{n-1} 2^m, \quad \text{then } q_N \geq m.$$

- (3) Using (2), deduce that the sequence $\{q_n\}_{n=1}^\infty$ diverges. Maybe you learned this in a calculus course?

Problem 5. Fix an integer $n > 0$.

- (1) Show that for any $\vec{u}, \vec{v} \in \mathbf{R}^n$, we have

$$| \|\vec{v}\| - \|\vec{u}\| | \leq \|\vec{v} - \vec{u}\|.$$

- (2) Using (1), show that the function $f: \mathbf{R}^n \rightarrow \mathbf{R}$ defined by $f(\vec{u}) = \|\vec{u}\|$ is analytically continuous.

Problem 6. Give an explicit function $f: \mathbf{R} \rightarrow \mathbf{R}$ with the given property. No need to justify your answer.

- (1) f is analytically continuous, but not bounded.
- (2) f is bounded, but not analytically continuous.
- (3) f is analytically continuous and bounded and has no maximum.
- (4) f is analytically continuous and bounded and $\lim_{x \rightarrow 0^+} f(1/x)$ does not exist.

Problem 7. Show that if $f, g: \mathbf{R}^n \rightarrow \mathbf{R}^m$ are both analytically continuous, then so is $f + g$.

Problem 8. Show that if $f: \mathbf{R}^n \rightarrow \mathbf{R}^m$ and $g: \mathbf{R}^m \rightarrow \mathbf{R}^\ell$ are analytically continuous, then so is $g \circ f$.

Problem 9. Show that any nonempty finite subset of $\mathbf{R}$ is analytically closed, but not analytically open.

Problem 10. Is the set of rational numbers analytically open in $\mathbf{R}$? Is it analytically closed?

Problem 11. Consider the following subset of $\mathbf{R}$:

$$K = \{\frac{1}{n} \mid n = 1, 2, 3, \dots\}.$$

Show that $K \cup \{0\}$ is analytically closed. Is K analytically closed?

Problem 12. View $\mathbf{R}$ as the x -axis in $\mathbf{R}^2$. In this way, we may view subsets of $\mathbf{R}$ as subsets of $\mathbf{R}^2$.

- (1) Show that no open subset of $\mathbf{R}$ remains analytically open in $\mathbf{R}^2$.
- (2) Nonetheless, show that any closed subset of $\mathbf{R}$ remains analytically closed in $\mathbf{R}^2$. *Hint:* Show that the complement $\mathbf{R}^2 - \mathbf{R}$ is open.

Problem 13. Give an example of an analytically continuous, non-constant map $f: \mathbf{R} \rightarrow \mathbf{R}$ and an analytically open set $U \subseteq \mathbf{R}$ such that $f(U)$ is *not* analytically open. *Hint:* Pick f to be polynomial.

Problem 14. Let X be a topological space, and let A be a subset of X . Suppose that for each $x \in A$, there is an open set U containing x such that $U \subseteq A$. Show that A is also open.

Problem 15. Show that an infinite intersection of analytically open sets in $\mathbf{R}$ need not be analytically open. Then repeat with “union” and “closed” in place of “intersection” and “open”.

Problem 16. Let X be any set. Show that the collection

$$\{\emptyset\} \cup \{U \subseteq X \mid X - U \text{ countable}\}$$

always forms a topology on X . What if we replace “countable” with “infinite” above?

Problem 17. Suppose that $X = \{a, b, c\}$ and

$$\mathcal{T}_1 = \{\emptyset, X, \{a\}, \{a, b\}\} \quad \text{and} \quad \mathcal{T}_2 = \{\emptyset, X, \{a\}, \{b, c\}\}.$$

Find the smallest topology containing $\mathcal{T}_1$ and $\mathcal{T}_2$ as subsets, and the largest topology that is contained in both $\mathcal{T}_1$ and $\mathcal{T}_2$ as a subset.

Problem 18. Given a topology $\mathcal{T}$, recall that a *basis* for $\mathcal{T}$ is a subcollection $\mathcal{B} \subseteq \mathcal{T}$ such that:

- (1) Every element of $\mathcal{T}$ is a union of elements of $\mathcal{B}$.
- (2) For any $U, U' \in \mathcal{B}$ and $x \in U' \cap U''$, there exists $U'' \in \mathcal{B}$ such that $x \in U''$ and $U'' \subseteq U' \cap U$.

Show that the countable set

$$\mathcal{B} = \{(a, b) \mid a < b \text{ and } a, b \text{ are rational}\}$$

forms a basis for the analytic topology on $\mathbf{R}$.

Problem 19. Let $a\mathbf{Z} + b = \{aq + b \mid q \in \mathbf{Z}\}$, for any integers a and b . Let

$$\mathcal{B} = \{a\mathbf{Z} + b \mid a, b \in \mathbf{Z} \text{ with } a \neq 0\}.$$

Show that $\mathcal{B}$ forms a basis for some topology on $\mathbf{Z}$. We refer to the topology it generates as the *evenly-spaced topology*. *Hint:* If $x \in a\mathbf{Z} + b$, then $a\mathbf{Z} + b = a\mathbf{Z} + x$.

Problem 20. Endow $\mathbf{Z}$ with the evenly-spaced topology from Problem 19.

- (1) Show that if there were finitely many (positive) primes, then $\{1, -1\}$ would be open. You may assume that any integer greater than 1 has a prime divisor.
- (2) Deduce that there must be infinitely many primes.